

PERFORMANCE ENHANCEMENT OF PLANT DISEASE DETECTION USING EFFICIENTNET-B0 WITH CBAM ATTENTION

AIML MINI PROJECT

PRESENTED BY:

AAKANKSHA CHAUDHARI - UCE2023610

SANIKA DEOKULE - UCE2023615

SRUSHTI JADHAV - UCE2023626

SAKSHI RATHOD - UCE2023655

GUIDE:

DR. MAHENDRA DEORE



INTRODUCTION

- Plant diseases significantly reduce crop yield & quality.
- Manual diagnosis → slow, requires expertise, not accessible to farmers.
- Deep Learning (CNNs) enables automated, fast, and high-accuracy diagnosis.
- EfficientNet models are lightweight and accurate.
- Attention mechanisms like CBAM help the model focus on disease-specific leaf regions.





WHY ACCURATE PLANT DISEASE DETECTION MATTERS?

- 40% crop loss happens globally due to plant diseases
- Farmers rely on manual diagnosis → slow, not always accurate
- Misdiagnosis leads to pesticide misuse & environmental harm



GOAL

- ✓ Detects diseases instantly from leaf images
 - ✓ Achieves 98–99% accuracy
 - ✓ Suggests remedies intelligently
 - ✓ Works for real-world farmers
- 



PROBLEM STATEMENT

01

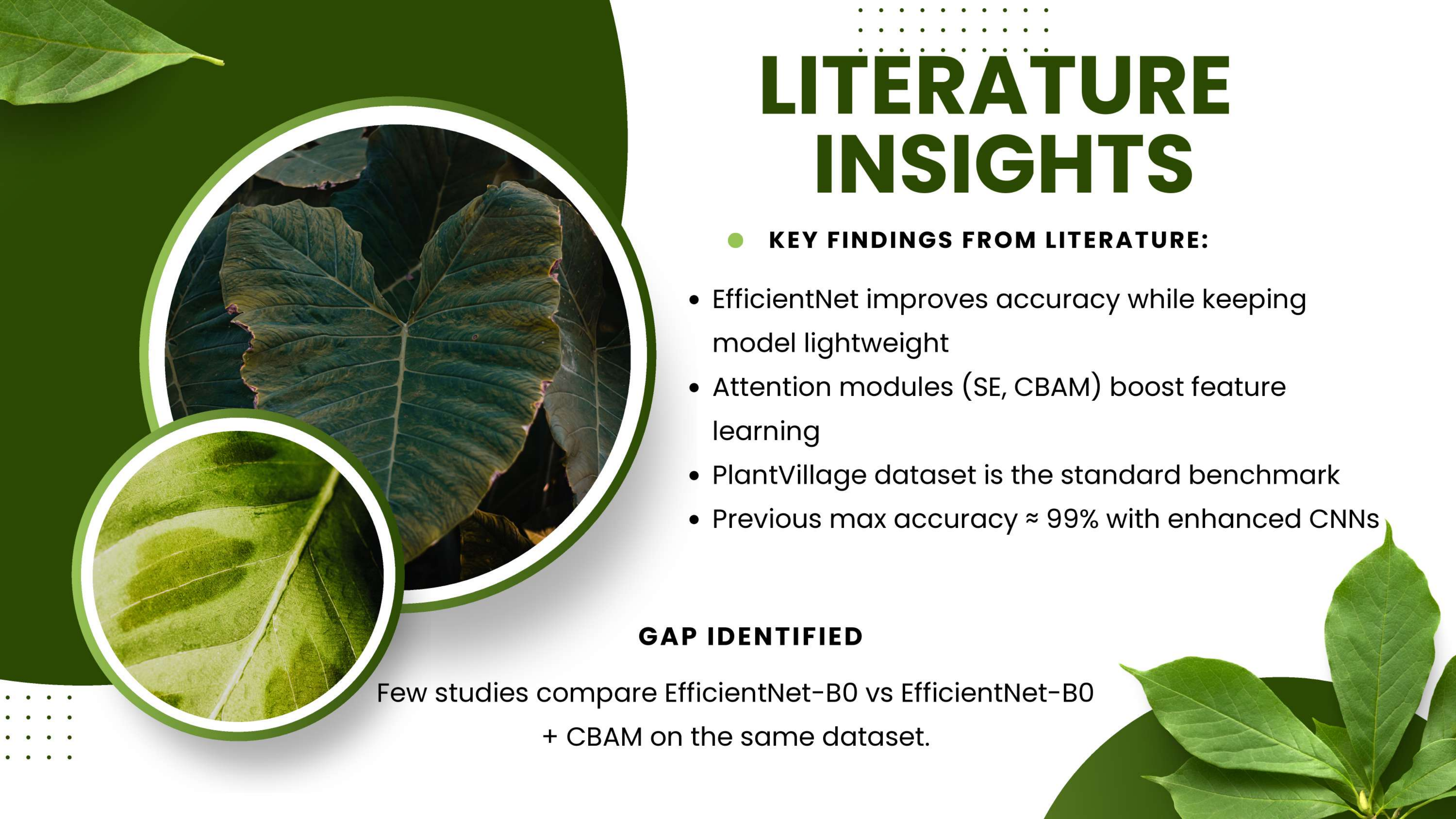
PROBLEM STATEMENT

1. Manual diagnosis is time-consuming
2. Requires agricultural experts
3. Symptoms of diseases look similar
4. Need for low-cost real-time detection

02

OBJECTIVES

- Detect plant diseases using EfficientNet-B0.
 - Improve accuracy using CBAM attention.
 - Validate results using PlantVillage dataset.
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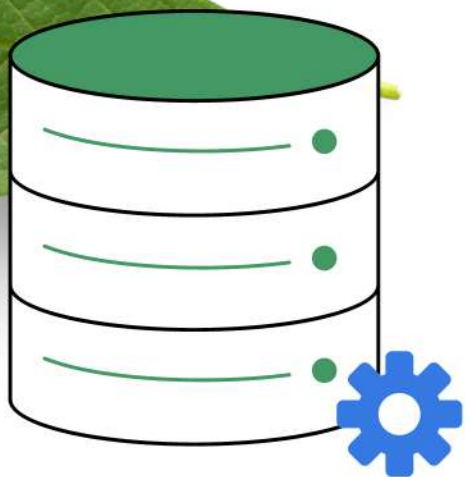
LITERATURE INSIGHTS

● KEY FINDINGS FROM LITERATURE:

- EfficientNet improves accuracy while keeping model lightweight
- Attention modules (SE, CBAM) boost feature learning
- PlantVillage dataset is the standard benchmark
- Previous max accuracy $\approx 99\%$ with enhanced CNNs

GAP IDENTIFIED

Few studies compare EfficientNet-B0 vs EfficientNet-B0 + CBAM on the same dataset.



DATASET OVERVIEW



Dataset: PlantVillage



Total Images: 54,000+



Classes: 38



Tomato, Potato, Apple, Grape, Pepper, etc.

Dataset Split:

- Training: 70%
- Validation: 15%
- Test: 15%

Preprocessing:

- Resize 224×224
- Normalization
- Augmentations (rotation, zoom, brightness, flips)





EFFICIENTNET-B0

01

BALANCED SCALING

EfficientNet introduces compound scaling, which increases:

- Depth (how many layers)
- Width (number of filters)
- Resolution (image size)

02

HIGH ACCURACY

EfficientNet-B0 achieves excellent accuracy even though it uses far fewer parameters than models like VGG or ResNet. @This makes the model faster and more efficient.

03

LIGHTWEIGHT & MOBILE-FRIENDLY

EfficientNet-B0 is optimized for real-time and mobile deployment, making it ideal for field use where farmers scan leaf images.





CBAM (CONVOLUTIONAL BLOCK ATTENTION MODULE)

Function

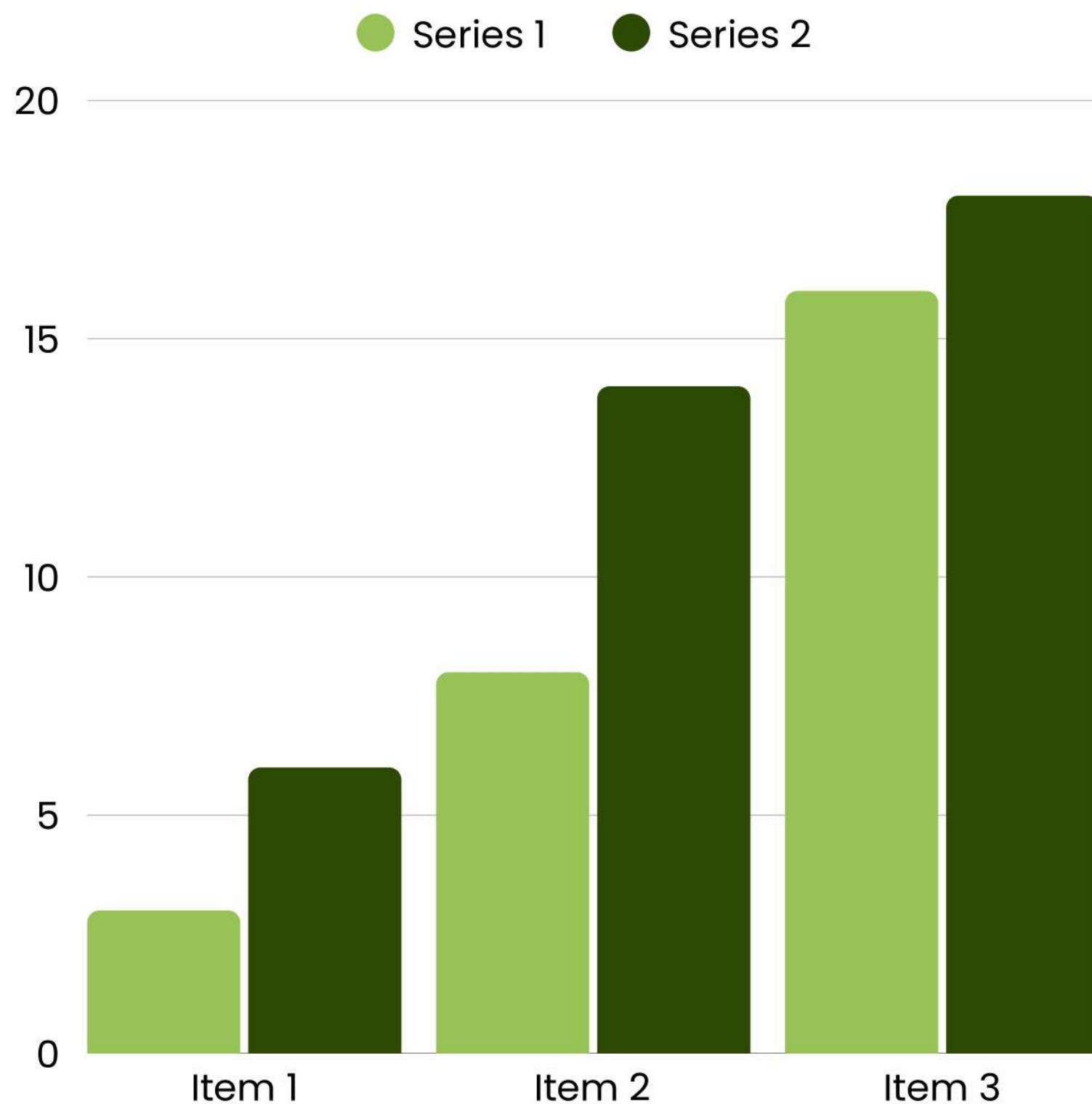
- the exact location of disease in the leaf
- highlights diseased patches
- ignores clean/healthy areas

Helps the model zoom in on infected regions.

Effect

- Focuses attention on the disease area
- Reduces noise and irrelevant background
- Improves feature quality
- Boosts classification accuracy

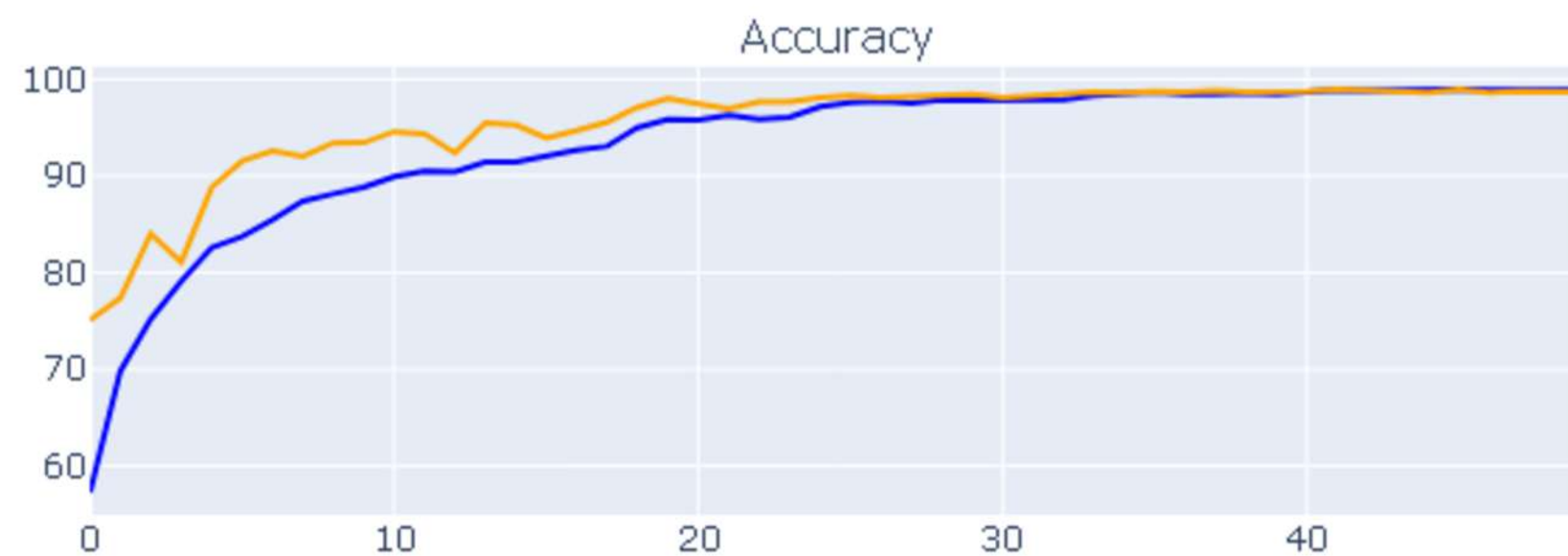
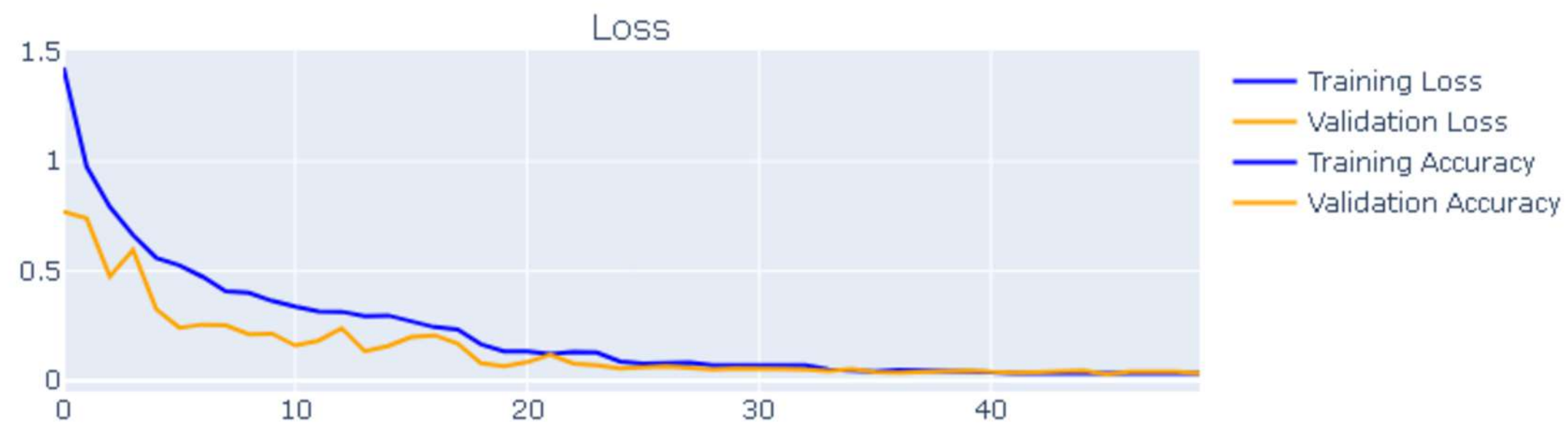




KEY OBSERVATIONS:

- ✓ Faster convergence
- ✓ Better feature focus
- ✓ Higher F1-score
- ✓ Better at visually similar diseases
- ✓ More stable validation accuracy

TRAINING MATRIX



CONCLUSION & FUTURE WORK

Conclusion

- CBAM significantly improved the model's performance.
- Achieved 99.42% accuracy, surpasses baseline.
- Attention mechanism helps to highlight disease-affected regions.
- Model is suitable for real-world implementation.



Future Scope

- ✓ Real-time mobile app for farmers
- ✓ IoT integration for smart farming
- ✓ Add Grad-CAM Explainability
- ✓ Extend dataset with field images
- ✓ Multi-language support

● TRANSFORMING DREAMS INTO REALITY

THANK YOU